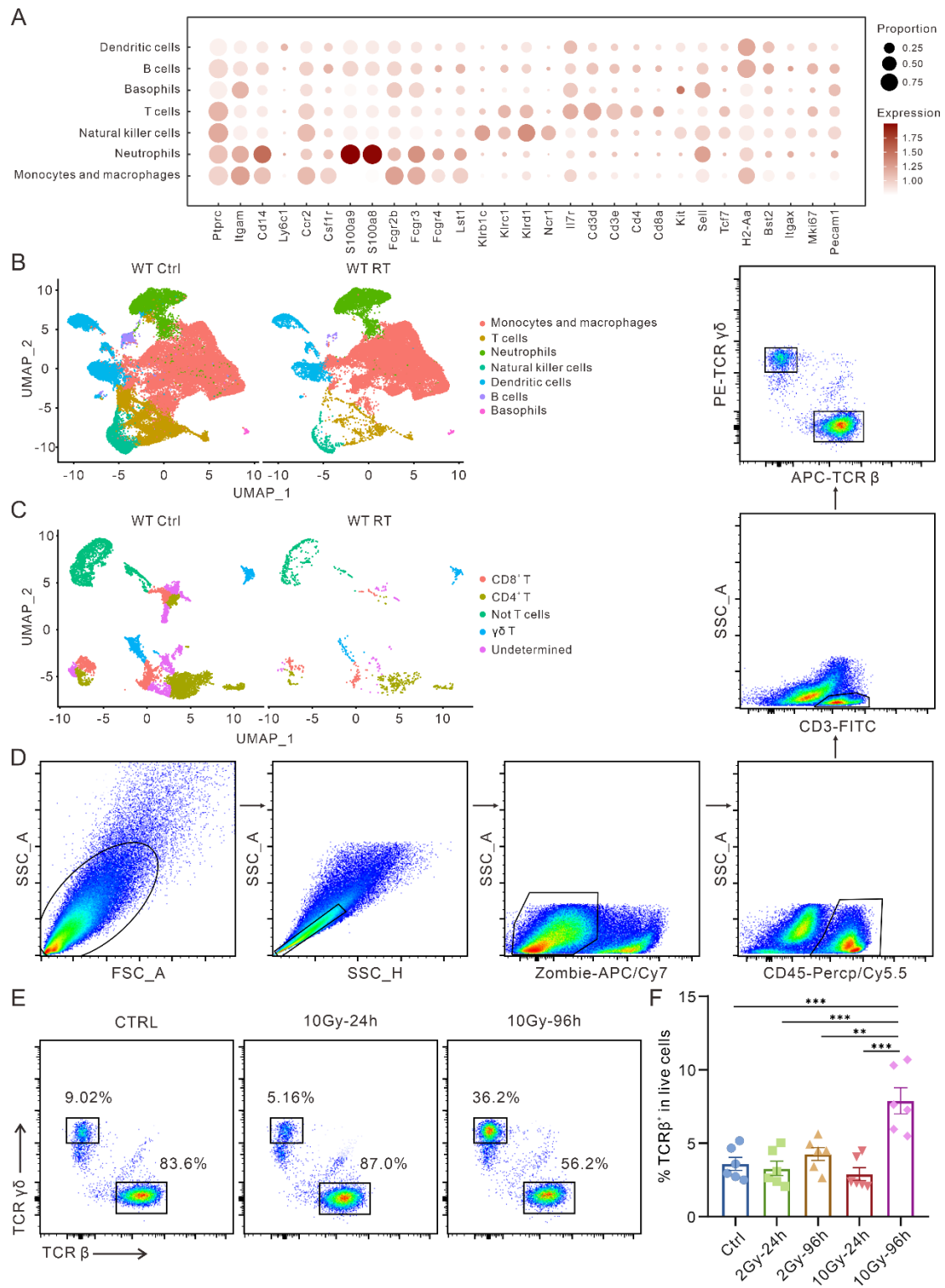


Supplemental material

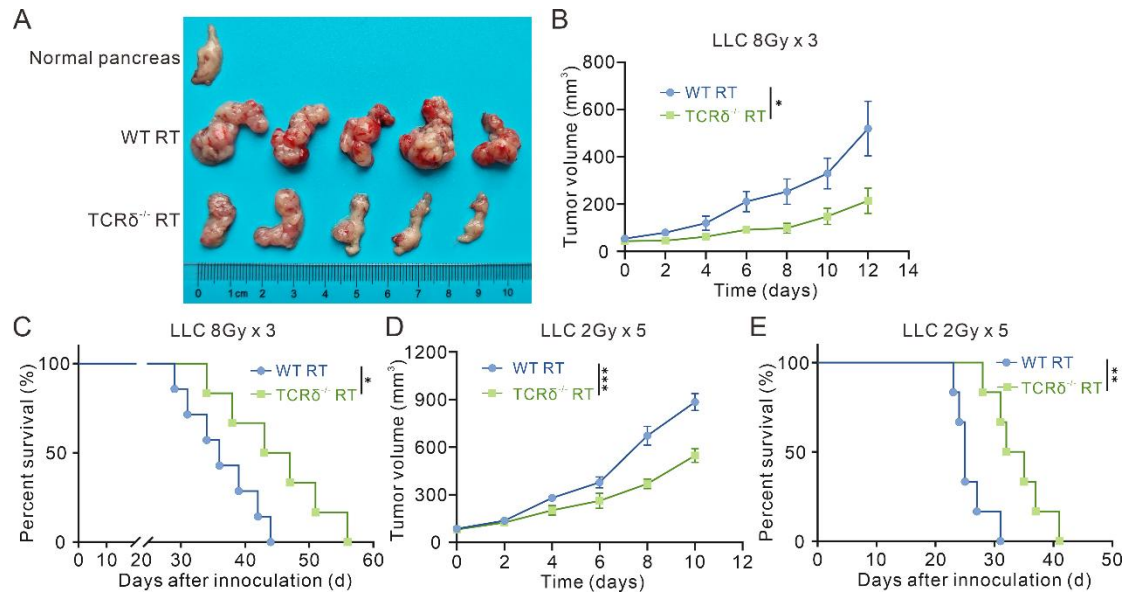
IL-17-producing $\gamma\delta$ T cells in the tumor microenvironment promote radioresistance in mice

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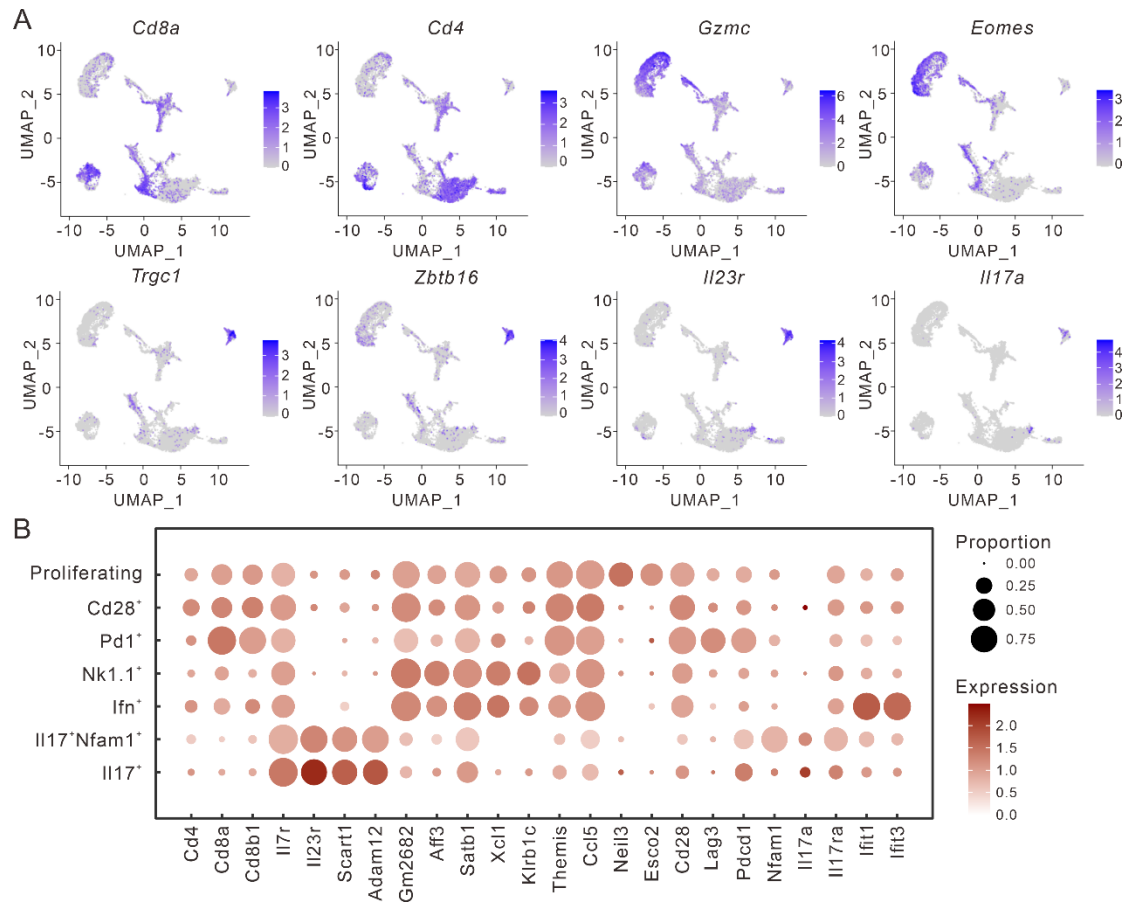


Supplemental Figure 1. Increased $\gamma\delta$ T cell infiltration in the TME after radiotherapy. (A) Dot plot showing the mean expression of canonical marker genes for seven major cell lineages identified by scRNA-seq. (B) UMAP plot of all cells from WT Ctrl and WT RT groups colored by cell identities. (C) UMAP plot of T cells from WT Ctrl and WT RT groups colored by cell clusters as indicated. (D) Flow cytometry gating strategy for identifying $\alpha\beta$ T cells and $\gamma\delta$ T cells. (E) Representative flow cytometry dot plots showing the percentages of $\alpha\beta$ T cells and $\gamma\delta$ T cells in the corresponding groups. (F) Flow cytometry analysis of $\alpha\beta$ T cell proportions in the TME of

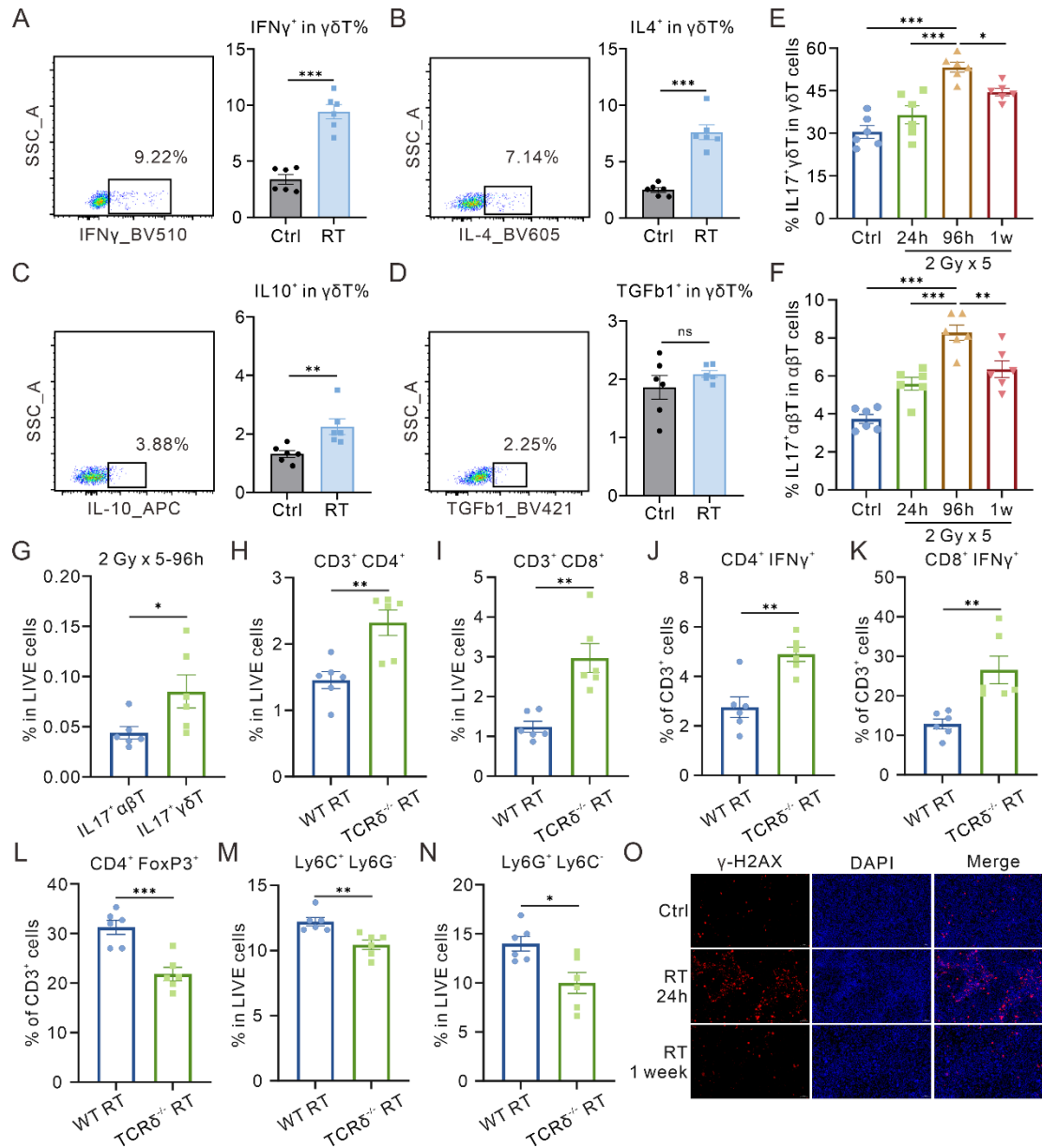
LLC subcutaneous tumors following radiotherapy at different doses (2 Gy and 10 Gy) and time points (24h and 96h) (n = 6 per group). ** $p < 0.01$; *** $p < 0.001$. One-way ANOVA with Tukey's multiple comparisons test (F).



Supplemental Figure 2. $\gamma\delta$ T cells drive resistance to radiotherapy. (A) Representative images of orthotopic pancreatic tumors and normal pancreas in the AKT/MYC-driven orthotopic model (n = 5 per group). (B) Tumor growth curves of LLC subcutaneous tumors following 8 Gy × 3 fractionated radiotherapy (n = 6 per group). (C) Kaplan-Meier survival plot of LLC lung cancer-bearing mice following 8 Gy × 3 radiotherapy (n = 6 to 7 per group). (D) Tumor growth curves of LLC subcutaneous tumors following 2 Gy × 5 fractionated radiotherapy (n = 5 to 6 per group). (E) Kaplan-Meier survival plot of LLC lung cancer-bearing mice following 2 Gy × 5 radiotherapy (n = 6 per group). * p < 0.05; ** p < 0.01; *** p < 0.001. Two-way ANOVA (B, D), Log-rank (Mantel-Cox) test (C, E).

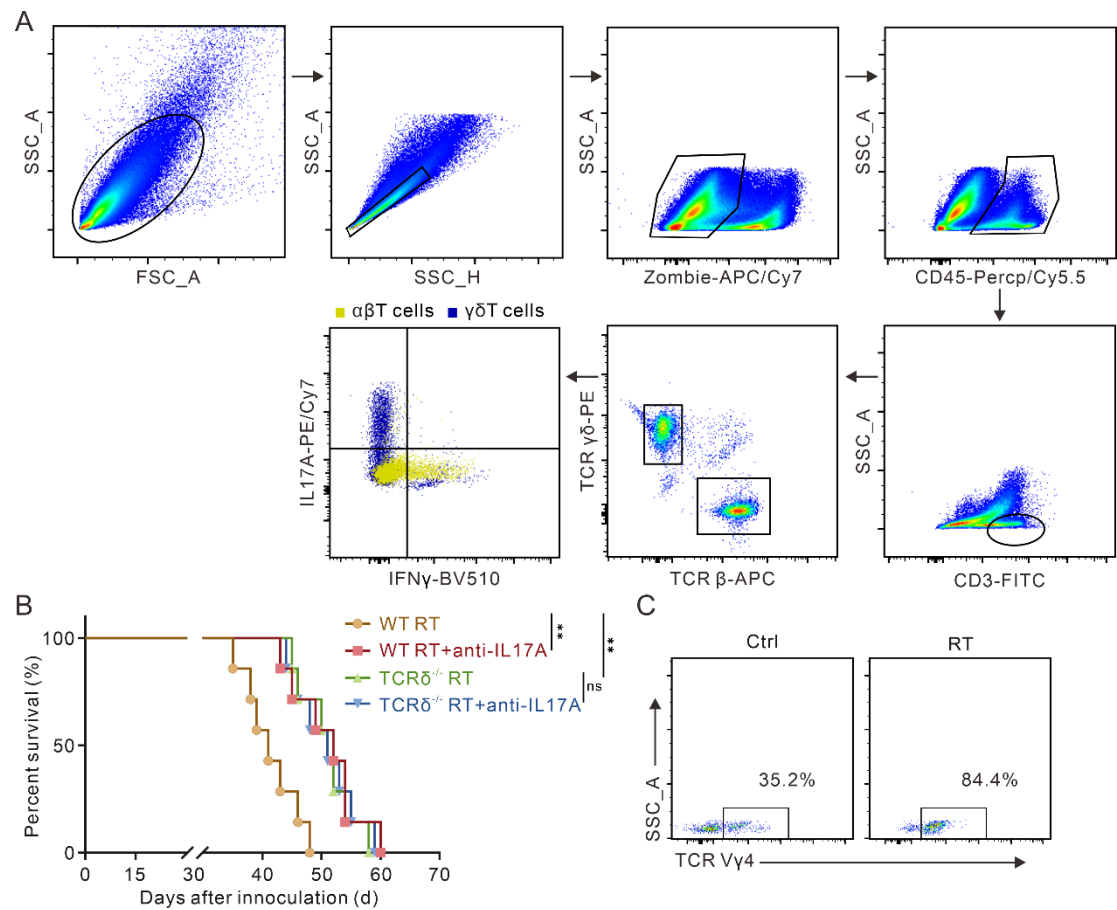


Supplemental Figure 3. scRNA-seq reveals the characteristics of T cells in the TME. (A) Feature plots depicting the expression of T cell characteristic genes in the major T cell clusters. **(B)** Dot plot showing the mean expression of canonical marker genes for seven major $\gamma\delta$ T cell clusters identified by scRNA-seq.



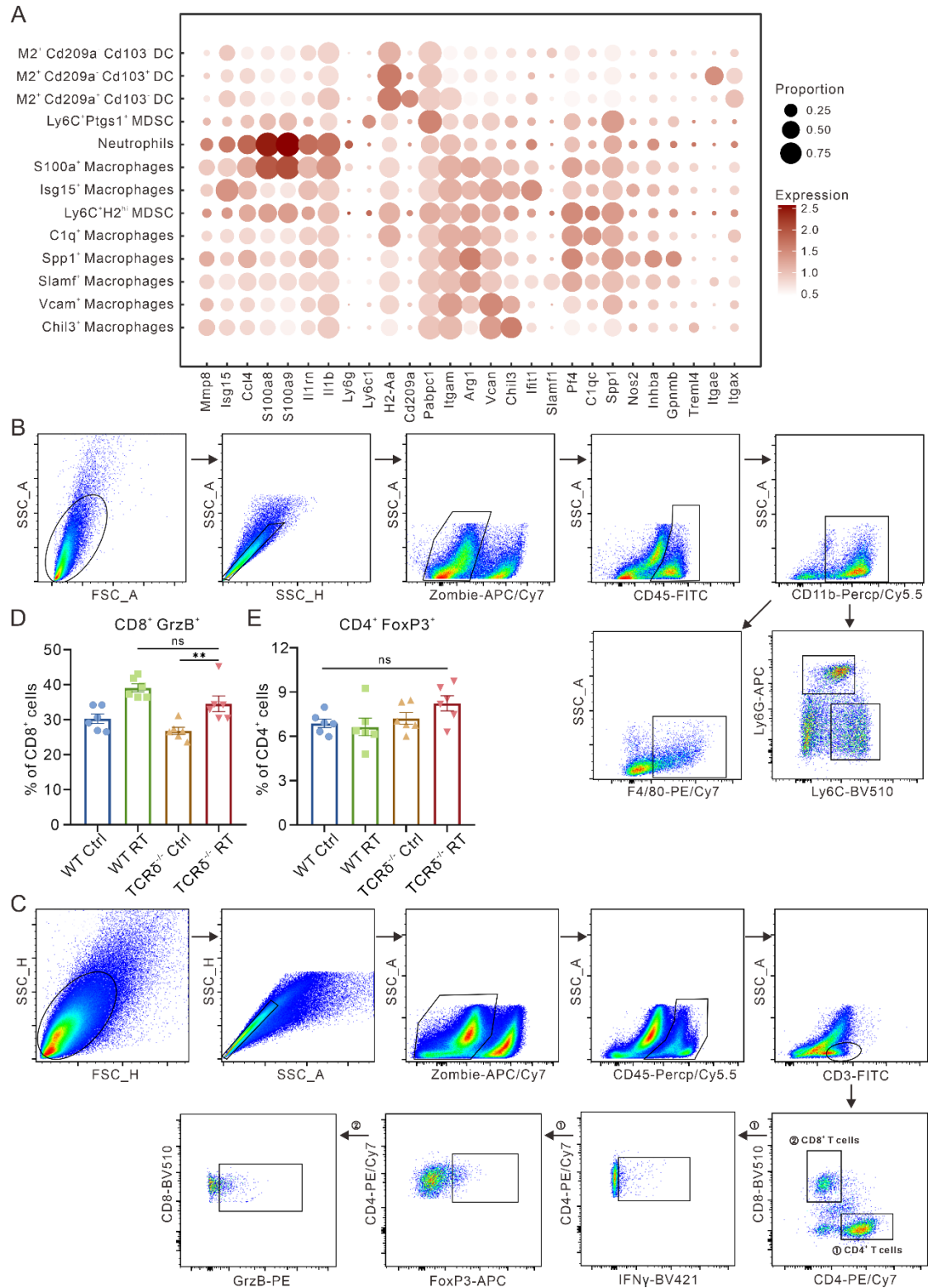
Supplemental Figure 4. Characteristics of $\gamma\delta$ T cells in the TME. (A-D) Representative flow cytometry plots and statistical analysis of IFN- γ (A), IL-4 (B), IL-10 (C), and TGF- β (D) expression in $\gamma\delta$ T cells from LLC subcutaneous tumors after radiotherapy (n = 6 per group). (E) Flow cytometry analysis of IL-17 $^+$ $\gamma\delta$ T cell proportions in the TME of LLC subcutaneous tumors following 2 Gy $\times$ 5 fractionated radiotherapy at different time points (n = 6 per group). (F) Flow cytometry analysis of IL-17 $^+$ $\alpha\beta$ T cell proportions in the TME of LLC subcutaneous tumors following 2 Gy $\times$ 5 fractionated radiotherapy at different time points (n = 6 per group). (G) Proportions of IL-17 $^+$ $\gamma\delta$ T cells and IL-17 $^+$ $\alpha\beta$ T cells in the TME at 96 hours after 2 Gy $\times$ 5 fractionated radiotherapy (n = 6 per group). (H-N) Flow cytometry analysis of CD3 $^+$ CD4 $^+$ T cell (H), CD3 $^+$ CD8 $^+$ T cell (I), CD4 $^+$ IFN γ^+ Th1 cell (J), CD8 $^+$ IFN γ^+ T cell (K), CD4 $^+$ FoxP3 $^+$ Treg cell (L), Ly6C $^+$ Ly6G $^+$ M-MDSC (M), and Ly6G $^+$ Ly6C $^+$ PMN-MDSC (N) proportions in the TME of LLC subcutaneous tumors at 1 week post-radiotherapy (n = 6 per group). (O) Representative γ -H2AX immunofluorescence staining of LLC subcutaneous tumor sections in corresponding

groups. $*p < 0.05$; $**p < 0.01$; $***p < 0.001$; ns, not statistically significant. Unpaired two-tailed Student's t-test (A-D, G-N), One-way ANOVA with Tukey's multiple comparisons test (E-F).



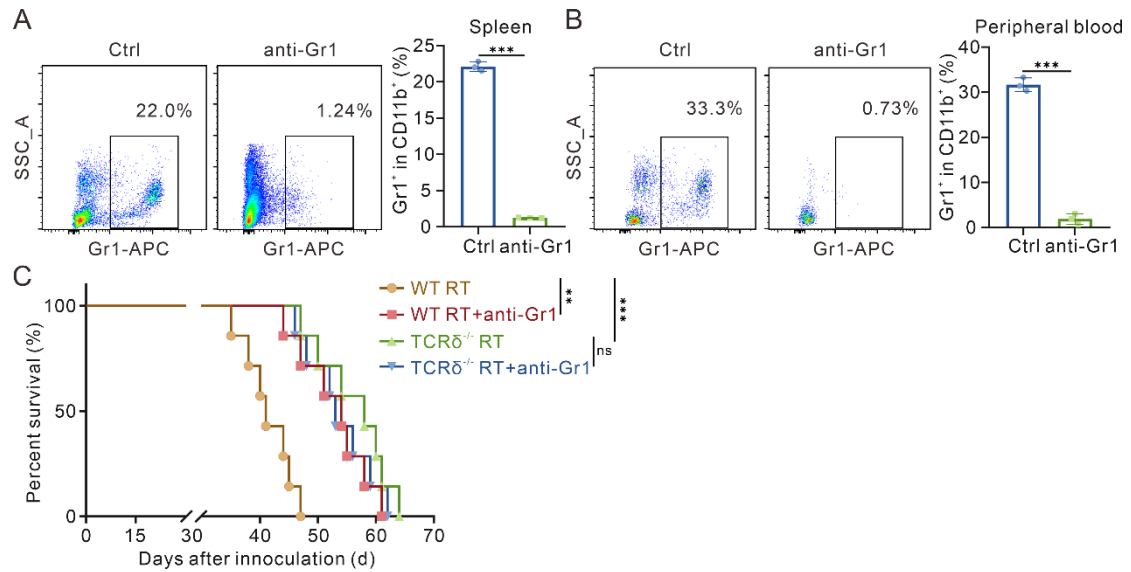
Supplemental Figure 5. $\gamma\delta$ T cells are the principal cellular source of IL-17A in the TME.

(A) Flow cytometry gating strategy for identifying cytokine production in $\alpha\beta$ T cells and $\gamma\delta$ T cells. (B) Kaplan-Meier survival plot of LLC lung cancer-bearing mice in corresponding groups. (n = 7 per group). (C) Representative flow cytometry plots of TCR V γ 4 $^{+}$ $\gamma\delta$ T cell proportions of LLC subcutaneous tumors after radiotherapy. ** $p < 0.01$; ns, not statistically significant. Log-rank (Mantel-Cox) test (B).

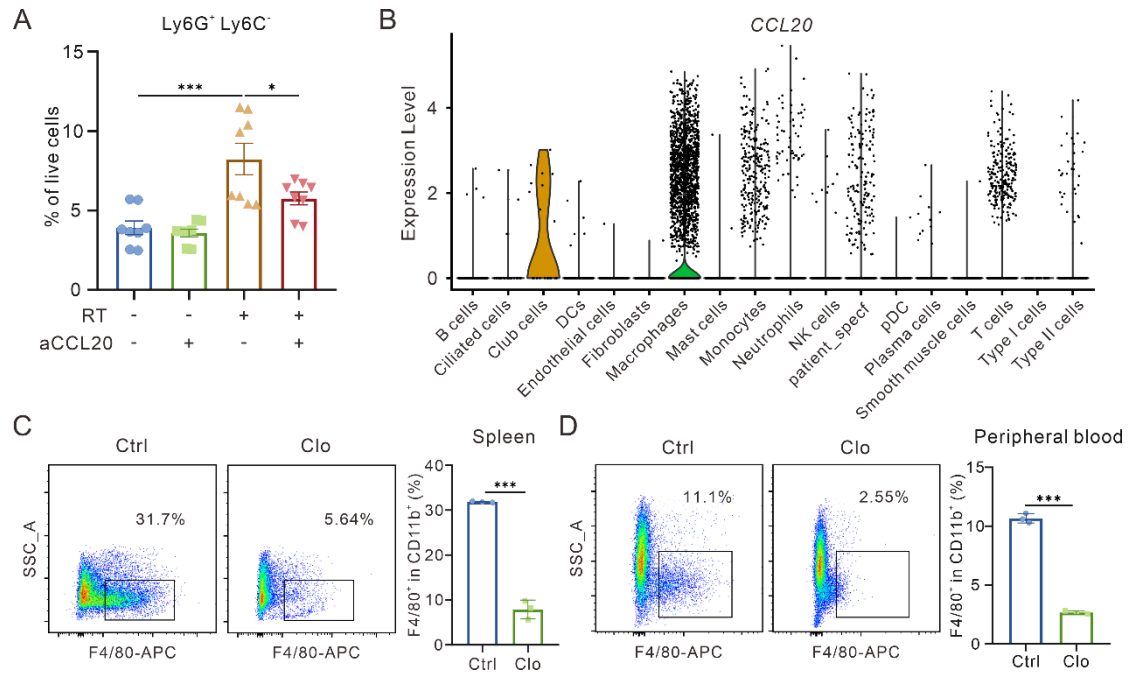


Supplemental Figure 6. $\gamma\delta$ T cells in the TME promote MDSCs recruitment and impair T cell activation. (A) Dot plot showing the mean expression of canonical monocytes and macrophages marker genes for major thirteen cell lineages identified by scRNA-seq. (B) Flow cytometry gating strategy for identifying myeloid cells. (C) Flow cytometry gating strategy for identifying CD4⁺ T cells and CD8⁺ T cells. (D-E) Flow cytometry analysis of CD8⁺GrzB⁺ T cell (D) and CD4⁺FoxP3⁺ T cell (E) proportions in the TME of LLC subcutaneous tumors in

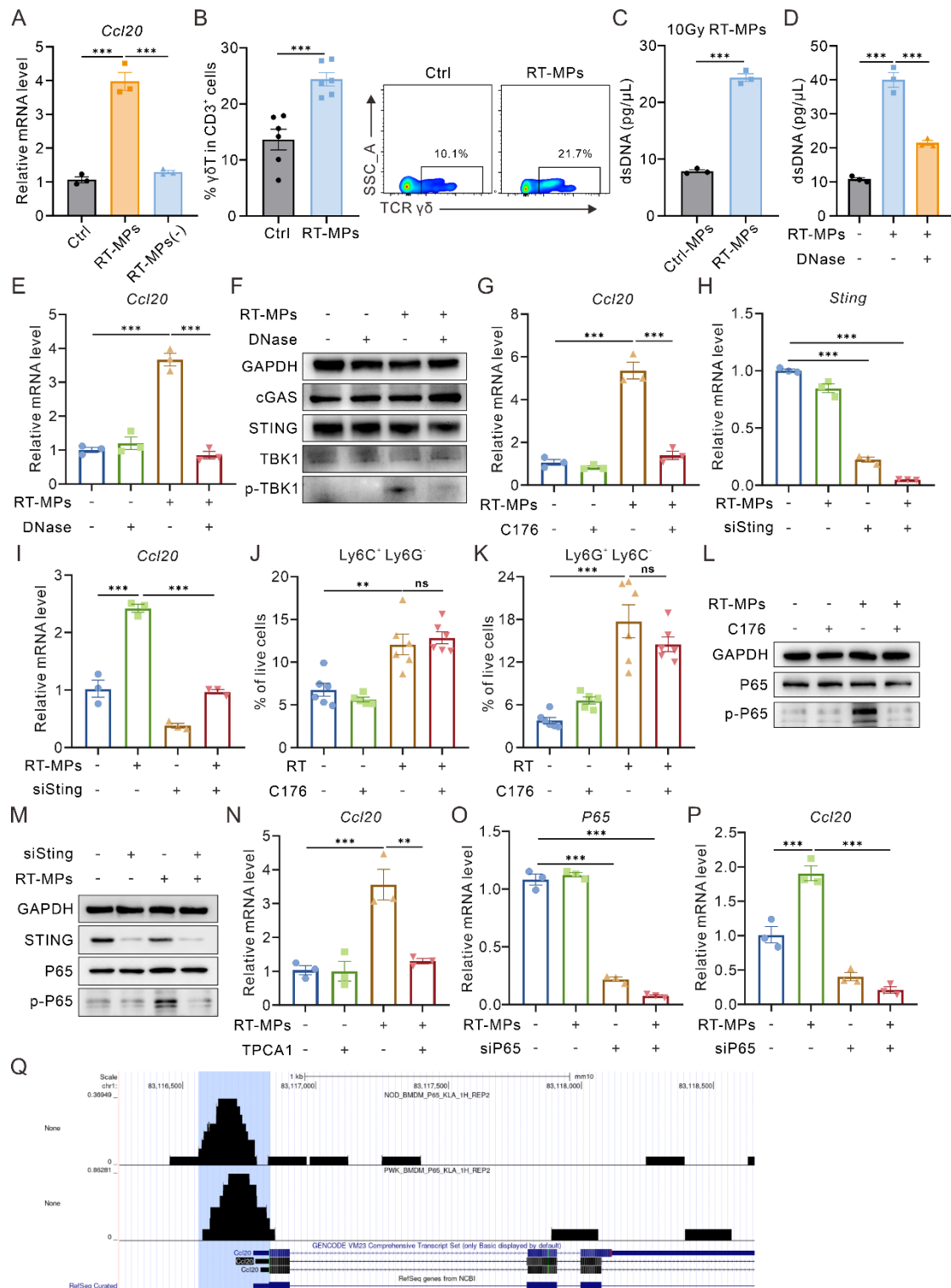
corresponding groups (n = 6 per group). $^{**}p < 0.01$; ns, not statistically significant. One-way ANOVA with Tukey's multiple comparisons test (D-E).



Supplemental Figure 7. $\gamma\delta$ T cells drive radioresistance via MDSC-mediated immunosuppression. (A) MDSCs clearance efficiency in mouse spleen detected by flow cytometry (n = 3 per group). (B) MDSCs clearance efficiency in mouse peripheral blood detected by flow cytometry (n = 3 per group). (C) Kaplan-Meier survival plot of LLC lung cancer-bearing mice in corresponding groups. (n = 7 per group). ** $p < 0.01$; *** $p < 0.001$; ns, not statistically significant. Unpaired two-tailed Student's t-test (A-B), Log-rank (Mantel-Cox) test (C).

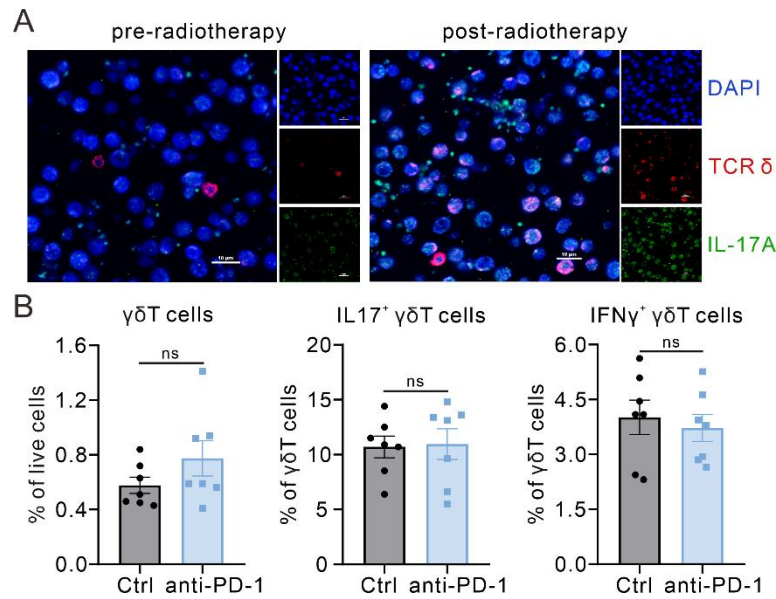


Supplemental Figure 8. Macrophages are the primary source of the chemokine CCL20. (A) Flow cytometry analysis of Ly6G⁺Ly6C⁻ PMN-MDSC proportions in the TME of LLC subcutaneous tumors in corresponding groups (n = 8 per group). **(B)** Violin plots demonstrate the *CCL20* expression in the major cell clusters. **(C)** Macrophages clearance efficiency in mouse spleen detected by flow cytometry (n = 3 per group). **(D)** Macrophage clearance efficiency in mouse peripheral blood detected by flow cytometry (n = 3 per group). **p* < 0.05; ****p* < 0.001. One-way ANOVA with Tukey's multiple comparisons test (A), Unpaired two-tailed Student's *t*-test (C-D).



Supplemental Figure 9. RT-MPs transcriptionally upregulate *Ccl20* expression in macrophages via the cGAS-STING/NF- κ B pathway. (A) Relative mRNA expression of *Ccl20* in BMDMs treated with 10 Gy-induced RT-MPs or irradiated tumor cell-derived CM depleted of RT-MPs. (B) Flow cytometry analysis of $\gamma\delta$ T cell proportions in the TME of LLC subcutaneous tumors after intratumoral injection of RT-MPs (n = 6 per group). (C) Quantitative measurement of dsDNA content in 10 Gy-induced RT-MPs and Ctrl-MPs. (D) Quantitative measurement of dsDNA content in RT-MPs treated with DNase. (E) Relative mRNA expression of *Ccl20* in

BMDMs treated with 10 Gy-induced RT-MPs or dsDNA-depleted RT-MPs. **(F)** Representative Western blot images showing protein expression levels of cGAS, STING, TBK1 and p-TBK1 in BMDMs treated with 10 Gy-induced RT-MPs or dsDNA-depleted RT-MPs. **(G)** Relative mRNA expression of *Ccl20* in BMDMs treated with 10 Gy-induced RT-MPs or the STING inhibitor C176. **(H)** Relative mRNA expression of *Sting* in BMDMs transfected with STING-targeting siRNAs. **(I)** Relative mRNA expression of *Ccl20* in BMDMs treated with RT-MPs or STING-targeting siRNAs. **(J-K)** Flow cytometry analysis of Ly6C⁺Ly6G⁻ M-MDSC (J), Ly6G⁺Ly6C⁻ PMN-MDSC (K) proportions in the TME of LLC subcutaneous tumors in corresponding groups (n = 5 to 6 per group). **(L)** Representative Western blot images showing protein expression levels of P65 and p-P65 in BMDMs treated with 10 Gy-induced RT-MPs or C176. **(M)** Representative Western blot images showing protein expression levels of STING, P65 and p-P65 in BMDMs treated with RT-MPs or STING-targeting siRNAs. **(N)** Relative mRNA expression of *Ccl20* in BMDMs treated with 10 Gy-induced RT-MPs or the NF-κB pathway inhibitor TPCA1. **(O)** Relative mRNA expression of *P65* in BMDMs transfected with P65-targeting siRNAs. **(P)** Relative mRNA expression of *Ccl20* in BMDMs treated with RT-MPs or P65-targeting siRNAs. **(Q)** UCSC Genome Browser tracks displaying the *Ccl20* promoter region with P65 ChIP-seq data from ENCODE. ***p* < 0.01; ****p* < 0.001; ns, not statistically significant. One-way ANOVA with Tukey's multiple comparisons test (A, D-E, G-K, N-P), Unpaired two-tailed Student's t-test (B-C).



Supplemental Figure 10. $\gamma\delta$ T cell depletion does not enhance the efficacy of immunotherapy. (A) Representative immunofluorescence staining of $\gamma\delta$ T cell marker TCR δ (red) and IL-17A (green) in peripheral blood PBMCs from lung cancer patients pre- and post-radiotherapy. Scale bar: 10 μ m. (B) Flow cytometry analysis of $\gamma\delta$ T cell, IL-17⁺ $\gamma\delta$ T cell, and IFN- γ ⁺ $\gamma\delta$ T cell proportions in the TME of LLC subcutaneous tumors after immunotherapy (n = 7 per group). ns, not statistically significant. Unpaired two-tailed Student's t-test (B).

Supplemental Table 1. Sequences of designed siRNA targeting *Sting* and *P65*.

siRNA	Sense Seq (5'-3')	Antisense Seq (5'-3')
<i>Sting</i>	GCCUCAUUGUCUACCAAGA	UCUUGGUAGACAAUGAGGC
<i>P65</i>	GCAGUUUGAUGCUGAUGAA	UUCAUCAGCAUCAAACUGC

Supplemental Table 2. Antibodies used in the Western blotting.

Antibodies	Source	Detailed information
GAPDH	ABclonal	GAPDH Rabbit pAb (AC001)
cGAS	Proteintech	cGAS Polyclonal antibody (29958-1-AP)
STING	Proteintech	STING Polyclonal antibody (19851-1-AP)
TBK1	Proteintech	TBK1 Polyclonal antibody (28397-1-AP)
Phospho-TBK1	CellSignalingTechnology	Phospho-TBK1/NAK (Ser172) Rabbit mAb (5483)
P65	ABclonal	NF- κ B p65/RelA Rabbit mAb (A19653)
Phospho-P65	Proteintech	Phospho-NF- κ B p65 (Ser468) Recombinant antibody (82335-1-RR)
Second antibody	ABclonal	HRP Goat Anti-Rabbit IgG (H+L) (AS014)

Supplemental Table 3. Sequences of primers for RT-qPCR.

Gene name	Forward Primer (5'-3')	Reverse Primer (5'-3')
<i>Gapdh</i> (m)	TGACCTCAACTACATGGTCTACA	CTTCCCATTCTCGGCCTTG
<i>GAPDH</i> (h)	CTGGGCTACACTGAGCACC	AAGTGGTCGTTGAGGGCAATG
<i>Ccl20</i> (m)	AGGCAGAAGCAAGCAACTACGAC	ATCGGCCATCTGTCTTGTGAAACC
<i>Sting</i> (m)	CCAAGAACCCACAGACGGAAACAG	GGAGGAGGTGCCACTGAGGTC
<i>P65</i> (m)	AGGCTTCTGGGCCTTATGTG	TGCTTCTCTCGCCAGGAATAC
<i>Ccl2</i> (m)	CACTCACCTGCTGCTACTCATTAC	CTTCTTTGGGACACCTGCTGCTG
<i>Ccl3</i> (m)	CTCCAGCCAGGTGTCATTTTCC	CAGGCATTCAGTTCCAGGTCAGTG
<i>Ccl4</i> (m)	TTGCTCGTGGCTGCCTTCTG	GAAGCTGCCGGGAGGTGTAAG
<i>Ccl5</i> (m)	GACACCACTCCCTGCTGCTTTG	CTCTGGGTGGCACACACTTGG
<i>Ccl8</i> (m)	GCTCCAGTCACCTGCTGCTTTC	ACACAGAGAGACATAACCCTGCTTGG
<i>Ccl17</i> (m)	GGTCACTTCAGATGCTGCTCCTG	CCTGGACAGTCAGAAACACGATGG
<i>Ccl19</i> (m)	TGCTAATGATGCGGAAGACTGCTG	TCCTTCTGGTGCTGTTGCCTTTG
<i>Ccl21a</i> (m)	GAGCCTCCTTAGCCTGGTCCTG	TGTTCACTTCTCTTGCAGCCCTTG
<i>Ccl22</i> (m)	CTCCTGGTGGCTCTCGTCCTTC	CGGCACAGATATCTCGGTTCTTGAC
<i>Ccl25</i> (m)	TTTGAAGACTGCTGCCTGGGTTAC	CACTCCTCACGCTTGTACTGTTGG
<i>Ccl27</i> (m)	CTACCGAAAGCCACTCTCAGACAAG	TCAGCCCATTTTCTTAGCATCCC
<i>Ccl27a</i> (m)	CTGGCATCCGTGGAACAAGACTAAG	GTCCCTTGGAGCCTTTTCCCTTG
<i>Ccl28</i> (m)	GTGTGTGTGGCTTTTCAAACCTCAG	AAGTACGATTGTGCGGGCTGATG
<i>Cxcl2</i> (m)	AACATCCAGAGCTTGAGTGTGACG	GGGCTTCAGGGTCAAGGCAAAC
<i>Cxcl5</i> (m)	TGCGTTGTGTTTGCTTAACCGTAAC	TGACTTCCACCGTAGGGCACTG
<i>Cxcl8</i> (m)	TCGGGAGACCTCTAGACACTTTGC	GCCTGTCAAGCTGACTTCACTGG
<i>Cxcl9</i> (m)	TCCTTTTGGGCATCATCTTCC	TTTGTAGTGGATCGTGCCTCG
<i>Cxcl10</i> (m)	CCAAGTGCTGCCGTCATTTTC	GGCTCGCAGGGATGATTTCAA
<i>Tcrvg4</i> (m)	CTTGCAACCCCTACCCATAT	AAATGTCTGCATCAAGTCT
<i>TRDV2</i> (h)	GCCATTGAGTTGGTGCCTGA	GGATGGTTTGGTATGAGGCTG
<i>TRGV9</i> (h)	GCAGGTCACCTAGAGCAACC	GAAGGAAGAAAAATAGTGGGC

Supplemental Table 4. Sequences of primers targeting *Ccl20* for ChIP.

	Forward Primer (5'-3')	Reverse Primer (5'-3')
Primer 1	CAGGAAGTTTTCCCTGTGGGT	GGGAATGTACACAAGAAGGCG
Primer 2	TTTGTGGTGACAGGATGAGGC	TCCAGCACCCAGCCCTTCTTAT